

Relations between HEXACO personality and ideology variables are mostly genetic in nature



Reinout E. de Vries¹ , Laura W. Wesseldijk¹,
Annika K. Karinen¹, Patrick Jern² and Joshua M. Tybur¹

Abstract

Existing work indicates that socio-political attitudes (or: ideology) are associated with personality, with Social Dominance Orientation and Right-Wing Authoritarianism relating most strongly to honesty-humility and openness to experience, the two value-related domains of the HEXACO framework. Using a sample of 7067 twins and siblings of twins (including 1376 complete twin pairs), we examined the degree to which these relations arise from common genetic and environmental sources. Heritability estimates for the HEXACO personality and ideology variables ranged from .34 to .58. Environmental factors shared by twins reared together showed negligible effects on individual differences in personality and ideology. At the phenotypic level, Social Dominance Orientation and Right-Wing Authoritarianism dimensions related most strongly to honesty-humility and openness to experience. These associations were mostly explained by genetic factors (48%–93%). Genetic correlations between openness to experience and the ideology scales ranged from $-.29$ to $-.53$; those between honesty-humility and the ideology scales ranged from $-.31$ to $-.43$. None of the environmental correlations exceeded $|r| = .18$. These results suggest that the relations between the two value-related domains of the HEXACO personality model and ideology are mostly genetic in nature, and that there is substantial overlap in the heritable components of personality and ideology.

Keywords

HEXACO, Social Dominance Orientation, Right-Wing Authoritarianism, heritability, personality

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Two ideological dimensions of socio-political attitudes—Social Dominance Orientation (SDO) and Right-Wing Authoritarianism (RWA)—form the bedrock of people's attitudes towards social groups, systems of government, the environment, social values, and numerous other societally relevant attitudes and behaviors (Claessens et al., 2020; Duckitt & Bizumic, 2013; Duckitt & Sibley, 2009; Dunwoody & Funke, 2016; Feather & McKee, 2012; Milfont et al., 2018; Pratto et al., 2006; Saucier, 2000; Stanley et al., 2017). The first socio-political attitude (hereafter called “ideology”) dimension, often referred to as Social Dominance Orientation, reflects preferences for group-based hierarchies. The second ideology dimension, often referred to as Right-Wing Authoritarianism, reflects adherence to traditional norms and values and willingness to submit to—and enforce—authoritarian rule. Both SDO and RWA covary with personality, with HEXACO honesty-humility the strongest correlate of SDO, and

openness to experience the strongest correlate of RWA (Desimoni & Leone, 2014; Lee et al., 2010; Leone et al., 2012a; Leone et al., 2012b). That is, to a considerable degree, ideology, as reflected in SDO and RWA, and personality, as reflected in honesty-humility and openness to experience, are intertwined.

While previous studies have established mostly robust heritability estimates of (HEXACO) personality (Kandler et al., 2019; Veselka, Schermer, et al., 2010) and the two ideology dimensions (Kandler et al., 2016; Kleppesø et al., 2019; Lewis & Bates,

¹Vrije Universiteit Amsterdam, Amsterdam, The Netherlands

²Åbo Akademi University, Turku, Finland

Corresponding author:

Reinout E de Vries, Institute for Brain and Behavior Amsterdam & Department of Experimental and Applied Psychology, Vrije Universiteit Amsterdam, Van der Boechorststraat 7, 1081 BT, Amsterdam, The Netherlands.

Email: re.de.vries@vu.nl

2014a), the degree to which relations between personality, SDO, and RWA arise from genetic versus environmental sources is currently unknown, as is the degree of overlap between the genetic and environmental factors underlying personality and ideology. In the current study, we used a twin design to: (1) estimate the phenotypic correlations between two recently developed measures of ideology, the SDO₇ (Ho et al., 2015) and the Aggression, Submission, and Conventionalism (ASC) scale (Dunwoody & Funke, 2016), which was designed to measure three facets of RWA; (2) estimate the heritability of these variables; (3) estimate the degree to which correlations between the variables are explained by genetic versus environmental factors; and (4) estimate the overlap in genetic factors (i.e., genetic correlations) and environmental factors (i.e., environmental correlations) that underlie these variables. We expected to observe phenotypic relations between personality and ideology similar to those observed in previous studies. Further, we expected to find that these relations are primarily undergirded by genetic factors. In sum, the current investigation aimed to answer the call to clarify “the underlying genetic and environmental associations between different characteristics [of personality, values, etc.]” (Kandler et al., 2014, p. 240).

Ideology and values

Research on socio-political attitudes (or: ideology) has often distinguished between two concepts: one pertaining to preferences for hierarchy and group discrimination (SDO) and the other to authoritarianism (RWA) (Claessens et al., 2020). SDO is defined as “a generalised orientation towards and desire for unequal and dominant/subordinate relations among salient social groups, regardless of whether this implies ingroup domination or subordination” (Pratto et al., 2006, p. 282), whereas RWA is defined in terms of three separate—but related—attitudes that support, strengthen, and propagate socio-political norms: authoritarian aggression, authoritarian submission, and conventionalism (Altemeyer, 1981).

Although separately conceptualized, SDO and RWA resemble generic conceptualizations of values and social attitudes. Schwartz (1992) has demonstrated that human values can be arranged in a circumplex spanned by two main dimensions: self-enhancement versus self-transcendence and conservation versus openness to change. Multiple studies have reported that SDO relates positively to self-enhancement (and negatively to self-transcendence) and RWA positively to conservation (and negatively to openness to change) (Altemeyer, 1998; Cohrs et al., 2005; Duriez & Van Hiel, 2002; Duriez et al., 2005; Feather & McKee, 2008, 2012). For instance, Feather and McKee (2012) found that SDO was most strongly

positively ($r = .45$) related to power, a value reflecting self-enhancement, and most strongly negatively ($r = -.47$) related to universalism, a value reflecting self-transcendence. In the same study, RWA was most strongly positively ($r = .44$) related to conformity, a value reflecting conservation, and most strongly negatively ($r = -.48$) related to self-direction, a value reflecting openness to change. Consequently, the two main value dimensions seem to be captured by SDO and RWA, respectively.

In the same vein, SDO and RWA relate distinctly to alphaism and betaism, the two most important dimensions in lexical research on societally relevant social beliefs (Saucier, 2000). In line with RWA and Schwartz' (1992) value dimension of conservation, the alphaism dimension reflected conservatism, traditionalism, and theism. In line with SDO and Schwartz' (1992) value dimension of self-enhancement, the betaism dimension reflected ethnocentrism, hedonism, and Machiavellianism. Consequently, SDO and RWA seem to represent the ideology domain as evidenced by their association with the two most important value- and social-attitudes dimensions.

Although SDO and RWA are often interpreted as singular (unidimensional) constructs (e.g., Lee et al., 2010; Osborne et al., 2017), researchers have argued that both should be conceptualized as multidimensional. Ho et al. (2015) distinguish SDO-Dominance (SDO-D) from SDO-Egalitarianism (SDO-E), stating that “the dominance dimension is characterized by support for overt oppression and aggressive intergroup behaviors designed to maintain the subordination of one or more groups, whereas the [anti-] egalitarianism dimension entails a preference for intergroup inequalities that are maintained by an interrelated network of subtle hierarchy-enhancing ideologies and social policies” (p. 1004). Although conceptualized as distinct, research has shown that SDO-D and SDO-E, as operationalized with the 16-item SDO₇ (Ho et al., 2015), are strongly related (i.e., $r > .70$ across three studies; Stanley et al., 2017). When correcting the three SDO₇ subscale correlations from Stanley et al.'s (2017) Study 2 and Study 3 (the latter which contains two waves of data) for attenuation due to unreliability, the average relation between the subscales is $r = .89$. Hence, SDO-D and SDO-E are virtually indistinct, at least as measured by the SDO₇.

With respect to RWA, Altemeyer (1981) suggested that RWA contains three distinctive item clusters, which he described as authoritarian aggression, authoritarian submission, and conventionalism. Duckitt and Bizumic (2013) conceptualized the RWA dimensional space using three subscales, respectively, authoritarianism, conservatism, and traditionalism (ACT), which differentially predicted attitudes toward different outgroups. However, the ACT scale has been criticized for its conflation of religiosity with traditionalism and for the inclusion of items that

are tautological with the criteria they pertain to predict (Dunwoody & Funke, 2016). To answer this criticism, Dunwoody and Funke (2016) reverted to Altemeyer's (1981) conceptualization of RWA by operationalizing Aggression, Submission, and Conventionalism (RWA-ASC) subscales of authoritarianism using a set of 18 newly minted items (containing three pro-trait and three con-trait items per subscale) that more narrowly and directly tap into the original definitions of these three constructs. Across two studies, the correlations between these subscales ranged from a low of .19 to a moderate .41, with a mean of .30. Furthermore, the subscales showed adequate discriminant validity in predicting a diverse set of political, social, and attitudinal criteria (Dunwoody & Funke, 2016). That is, in contrast to the SDO₇, the RWA-ASC subscales seem to tap sufficiently distinct content.

HEXACO traits, values, and ideology

As noted above, SDO and RWA share a considerable amount of variance with values. As we will show, personality is also related to values and, consequently, to ideology. Lexical research has shown that personality can be captured by a maximum set of six cross-culturally replicable factors of personality (Ashton et al., 2004; De Raad et al., 2014; Saucier, 2009), represented by the personality dimensions that form the HEXACO acronym, i.e., Honesty-humility, Emotionality, eXtraversion, Agreeableness, Conscientiousness, and Openness to experience (Lee & Ashton, 2004). Of these six factors, honesty-humility is most strongly associated with Schwartz' value dimension of self-enhancement versus self-transcendence and SDO, and openness to experience is most strongly associated with Schwartz' value dimension of conservation versus openness to change and RWA (Desimoni & Leone, 2014; Lee et al., 2009; Lee et al., 2010; Leone et al., 2012a; Leone et al., 2012b; Pozzebon & Ashton, 2009). Note that Big Five agreeableness, which captures some variance associated with honesty-humility (see Ashton & Lee, 2020, for an overview of differences between Big Five and HEXACO models), most strongly relates to SDO (Ho et al., 2015). In the HEXACO model, agreeableness is much less strongly related to both Schwartz' self-transcendence dimension and SDO (Lee et al., 2009; Leone et al., 2012a; Leone et al., 2012b; Ho et al., 2015).¹

Of all six HEXACO dimensions, honesty-humility and openness to experience appear to be the most consistent ideology- and value-related personality dimensions. That is, these two personality dimensions represent the relatively stable differences among individuals in their typical affective, behavioral, and cognitive responses to situations in which ideology- and value-based assessments or choices are—or must be—made. With respect to honesty-humility, such

assessments and choices pertain to situations that allow for exploitation (versus cooperation for the public good), in which self-enhancing versus self-transcendent choices can be made that may—or may not—result in sexual, material, and/or status gains. According to the Situation-Trait-Outcome Activation (STOA) model (De Vries et al., 2016), people low on honesty-humility are more likely to select situations (situation activation) that may offer them self-enhancing (sexual, material, and/or status) benefits (Julian et al., 2021; Lee et al., 2013). In such situations that allow for exploitation, different levels of honesty-humility are more likely to be expressed and noticeable (trait activation), and, as a consequence, low (or high) honesty-humility can be more or less successful (outcome activation). For instance, in high power (status) situations, in which exploitation of others is possible, low honesty-humility results in higher monetary gains (Barends et al., 2019). Because of the preference for situations that allow for exploitation, people low in honesty-humility, like people high in SDO, are more likely to prefer situations in which other people—and social groups—can be subordinated and oppressed.

With respect to openness to experience, such ideology- and value-based assessments and choices pertain to situations that allow for exploration (versus rejection of change), in which curiosity and the investigation of new and unconventional ideas can lead to personal, group, and societal benefits or harm. According to the STOA model, people who are high on openness to experience are more likely to select situations and explore ideas that may run counter to conventional societal norms. In such situations, openness to experience is activated. This activation can lead to positive or negative personal, group, and societal outcomes (De Vries et al., 2016). For instance, because openness to experience is ridiculed in traditional, low openness to experience communities (Camperio Ciani & Capiluppi, 2011; Mueller et al., 2012), people who are high on openness to experience are more likely to migrate to environments that are open to new and unconventional ideas (Camperio Ciani & Capiluppi, 2011; Chen et al., 1999; Matthews & Butler, 2011). Because of their preference for situations that allow for exploration, people scoring high on openness to experience, like people low on RWA, do not like to submit themselves to conventions, authorities, or authoritarian rule.

The reasoning outlined above raises a key question: are personality and ideology causally related? Because personality is generally believed to be present early in life, whereas ideological attitudes are believed to develop in (late) adolescence (Altemeyer, 1998; Sibley & Duckitt, 2013), the causality argument implies that personality influences ideology, and not—or at least not as strongly—vice versa. In terms of the STOA model (De Vries et al., 2016), this reasoning would imply that personality has an

effect on situation selection (e.g., the peer environment) and the outcomes (positive or negative) one experiences in such environments, which in turn influence the ideological expressions one is exposed to and the extent to which such attitudes are rewarded or punished. Longitudinal studies (Perry & Sibley, 2012; Sibley & Duckitt, 2013) have hinted at a causal relation between personality and ideology, with personality prospectively predicting ideology to a greater extent than the reverse. However, a twin study on personality and ideology, using the Eysenck Personality Questionnaire (Eysenck & Eysenck, 1975) and a direction-of-causation model, concluded that there is not much evidence for a causal relation from personality to ideology and thus that they are more likely to share a common origin (Verhulst et al., 2012; see also Kandler et al., 2014). Yet, no work has examined the underlying genetic and/or environment sources and genetic correlations (i.e., overlap in genetic influences) between recent conceptions of personality and ideology (i.e., HEXACO, SDO₇, and RWA-ASC).

Heritability of HEXACO personality and ideology variables

Meta-analyses of twin studies have shown heritability estimates of .48 for Big Five personality traits (Briley & Tucker-Drob, 2014; Vukasović & Bratko, 2015), though estimates derived from family and adoption studies are lower (e.g., .19; Vukasović & Bratko, 2015). The only two studies that have investigated the heritability of the HEXACO personality traits have found similar average heritabilities (e.g., an unweighted mean of .48; Kandler et al., 2019; Veselka, Schermer, et al., 2010).² In these HEXACO and Big Five twin studies, environmental factors shared by family members, such as twins reared together, have been found to have a negligible impact on individual differences in personality, whereas unique environmental effects (and error of measurement) not shared by family members have been found to account for the remaining variance.

Additional genetically informed investigations are especially warranted for the more recent HEXACO model because of some notable differences from the Big Five model. The HEXACO model is isomorphic to the Big Five model with respect to extraversion, conscientiousness, and openness to experience, three traits that are theorized to represent engagement with respectively the social, task, and ideas domain. When compared to the Big Five model, the HEXACO honesty-humility, emotionality, and agreeableness factors contain both additional variance (mainly honesty-humility) and redistributed variance (mainly emotionality and agreeableness) when compared with the Big Five emotional stability and agreeableness factors (Ashton et al., 2014). Consequently, although one would expect heritabilities of HEXACO

emotionality and agreeableness to be reasonably similar to those observed for Big Five emotional stability and agreeableness, the same may not be true for honesty-humility. In both Kandler et al. (2019) and Veselka, Schermer, et al. (2010), heritabilities of honesty-humility (respectively .46 and .35) were somewhat (~.10) lower than the average heritability of the other five traits (respectively .56 and .44). That is, the dearth of genetically informed studies using the HEXACO model, combined with evidence that honesty-humility changes more during the lifetime than any other personality factor (Ashton & Lee, 2016; Thielmann & De Vries, 2021)—implying that it is more “malleable” than other factors—makes an investigation of the heritability of the HEXACO factors especially worthwhile.

Twin studies on RWA report heritabilities similar to those found for personality (e.g., between .47 and .50 (Funk et al., 2013; Lewis & Bates, 2014a, Kandler et al., 2016; McCourt et al., 1999). Heritability estimates for SDO have been lower (respectively .37 and .24 for SDO-D and SDO-E in Kleppstø et al. (2019) and .02 in Kandler et al. (2016)).³ In contrast to the lack of shared environmental effects in personality, researchers have found that environmental factors shared by relatives influence both SDO and RWA (in univariate twin modelling results, respectively .32 for SDO and .24 for RWA in Kandler et al. (2016), .16 for RWA in McCourt et al. (1999), and .12 for RWA in Funk et al. (2013)). Because especially honesty-humility and SDO have been found to have (somewhat) lower heritabilities than the other personality and ideology traits, understanding whether the phenotypic correlation between the two is predominantly genetic or environmental is especially important. The current study aims to address this issue.⁴

The present study

The main goal of the present study was to test the phenotypic relations between HEXACO personality traits and the newly developed measures of SDO and RWA and to examine the degree to which these relations are undergirded by shared and unique genetic and/or environmental factors. We laid out three additional secondary goals. First, our study afforded the opportunity to test the factor structure of recently developed measures of SDO (the SDO₇; Ho et al., 2015) and RWA (the RWA-ASC scale; Dunwoody & Funke, 2016). Two- (SDO) and three- (RWA) factor structures would be expected, but doubts remained given the high correlation between the two SDO subscales (Stanley et al., 2017). Second, our study allowed us to estimate the heritability of HEXACO personality and ideology variables, which is especially important given inconsistencies in findings regarding the heritability of ideology (Kandler et al., 2016; Kleppstø et al., 2019). Third, our study allowed us to estimate the extent to which

relations between HEXACO traits and ideology reflect genetic and environmental sources. Given that twin research on personality rarely reveals effects of the shared environment (Briley & Tucker-Drob, 2014), we did not expect that relations between personality and ideology would arise from the shared environment. However, the low heritability and non-zero shared environmental effects of SDO and the somewhat lower heritability of honesty-humility may suggest that the typical phenotypic relation observed between honesty-humility and SDO may have an unshared environmental origin. That is, this research can reveal the extent to which dimensions of personality and ideology arise from genetic and/or environmental factors and to what degree those factors overlap.

Method

Data, analysis scripts, supplemental information, and supplemental tables are provided on <https://osf.io/yh xwt/>.

Sample and procedure

The data collection plan was approved by the Ethical Review Board of Åbo Akademi University and sampling information is provided in Tybur et al. (2020). We invited 33,211 twins and siblings of twins drawn from Finland's Central Population Registry to participate in this study, which was conducted online. A maximum of two reminder letters were sent to participants who did not respond to the initial letter, spaced two to three weeks apart. Data collection began in November 2018 and concluded in January 2019. In total, 9319 participants completed at least part of the survey and gave consent for the use of their data for scientific purposes. For these analyses, we only included participants who reported their sex and age and completed the 100-item version of the HEXACO Personality Inventory-Revised (HEXACO-100; $N = 6190$ twins and $N = 877$ siblings). This sample consisted of 1376 complete twin pairs (132 monozygotic male (MZm), 407 monozygotic female (MZf), 118 dizygotic male (DZm), 360 dizygotic female (DZf), and 359 dizygotic opposite-sex (DOS)). Out of the 1376 twin pairs, 1353 had data available both on the HEXACO-100 and SDO and 1354 on HEXACO-100 and RWA-ASC. Twins were aged between 18 and 45 years ($M = 29.28$, $SD = 7.75$) and siblings between 18 and 58 ($M = 31.99$, $SD = 8.43$). See Tybur et al. (2020) for a more detailed description of data collection procedures and zygosity estimation.⁵

Instruments

HEXACO-100. The HEXACO-100 (Lee & Ashton, 2018) includes 16 items for each of the six HEXACO domain scales that form the HEXACO

acronym: (1) Honesty-humility, (2) Emotionality, (3) eXtraversion, (4) Agreeableness, (5) Conscientiousness, and (6) Openness to experience. Multigroup confirmatory factor analyses of the HEXACO-100 in 16 languages have shown that the factor structure and the meaning of the HEXACO domains are comparable across languages (Thielmann et al., 2020). Four four-item facet subscales are embedded within each domain, and an interstitial altruism facet loads on multiple domains. Items on this latter facet are not included in the current analyses. Each item is answered on a 1- to 5-point scale (1 = strongly disagree, 2 = disagree, 3 = neutral (neither agree nor disagree), 4 = agree, and 5 = strongly agree). After reverse coding the reverse-keyed items (between 7 and 10 per factor), we took the mean of the 16 items per factor. McDonald's omega was adequate for each (.82, .84, .90, .82, .79, and .83 for H, E, X, A, C, and O, respectively).⁶

Social Dominance Orientation. The SDO₇ (Ho et al., 2015) includes 16 items, 8 of which are designed to measure dominance (e.g., "Some groups are simply inferior to other groups") and 8 of which are designed to measure anti-egalitarianism (e.g., "It is unjust to try to make groups equal"). All items were endorsed on a 7-point scale ranging from 1 = strongly oppose to 7 = strongly favor.

Right-Wing Authoritarianism. The Aggression-Submission-Conventionalism RWA questionnaire (RWA-ASC; Dunwoody & Funke, 2016) is a recently developed 18-item instrument designed to capture the three dimensions of RWA proposed by Altemeyer (1981): authoritarian aggression, authoritarian submission, and conventionalism (hereafter referred to as RWA aggression, RWA submission, and RWA conventionalism). This instrument was developed to improve upon two shortcomings of existing RWA measures. First, it aimed to avoid earlier measures' double-barreled item formats, which obscured the ability to separately estimate the three presumably distinct factors. Second, it aimed to avoid items with content relating to religion or prejudice, given that RWA instruments are often used to predict these constructs. For example, whereas other approaches to measuring conventionalism include items like "God's laws about abortion, pornography, and marriage must be followed before it is too late," RWA conventionalism items include no religious content and instead focuses on traditions and norms (e.g., "Traditions are the foundation of a healthy society and should be respected"). These modifications are especially important for measurement across nations that do not widely follow Abrahamic religions.

Analytic approach

As detailed in the Supplemental Information file, we performed eight different confirmatory factor analyses (CFAs) on the ideology items. In all eight models, reverse-scored items loaded on a latent social desirability factor. In model 1a, apart from the social desirability factor, we modeled a single SDO latent variable and three RWA latent variables, each of which corresponded with one of the ASC scales. This model, which did not fit well ($\chi^2(504) = 24,342.91$, $p < .001$, root mean square error of approximation (RMSEA) = .08, comparative fit index (CFI) = .81, Tucker-Lewis index (TLI) = .79), was improved by adding selected error covariances between items loading on the same latent variable (model 1b; $\chi^2(486) = 11,397.63$, $p < .001$, RMSEA = .05, CFI = .92, TLI = .91; $\Delta\chi^2(18) = 12945$, $p < .001$). In model 2a, we modeled separate latent variables for SDO-E and SDO-D as well as the three RWA-ASC scales. Again, this model did not fit well ($\chi^2(500) = 18,778.58$, $p < .001$, RMSEA = .07, CFI = .86, TLI = .84), though again model fit was improved by adding selected error covariances (model 2b; $\chi^2(482) = 8685.41$, $p < .001$, RMSEA = .05, CFI = .94, TLI = .93; $\Delta\chi^2(18) = 10093$, $p < .001$). All four models were rerun, adding a higher-order factor for RWA (models 1a and 1b) and higher-order factors for both RWA and SDO (models 2a and 2b). These models did fit the data better than models that did not include higher-order factors. Although model 2b fit better than model 1b, the correlation between the two SDO latent variables was sufficiently high (.82) (see also Stanley et al., 2017) that we proceeded with computing factor scores based on item loadings from model 1b. McDonald's omega reliabilities for the four factor scales were .80 (SDO), .70 (RWA aggression), .70 (RWA submission), and .76 (RWA conventionalism).

For each of the 10 variables (6 HEXACO, 1 SDO, and 3 RWA), we used structural equation modeling in OpenMx (Boker et al., 2011) to examine correlations within twin pairs for MZm, DZm, MZf, DZf, and DOS twins, as well as correlations between the male and female sibling with their twin sibling. Next, raw scores were age-corrected by regressing them on age and saving residualized values. Correlations between the age-corrected variables were submitted to a genetic structural equation model, in which phenotypic variance (based on within-trait correlations between twin 1 and twin 2) is partitioned into additive genetic (A), either common environmental (C) or non-additive dominant genetic (D), and non-shared environmental (E) components. This method assumes that monozygotic (MZ) twins share (close to) 100% of their segregating genetic material and common environment (including being raised by the same parents, having the same childhood socioeconomic status, etc.), whereas dizygotic (DZ) twins and

siblings share, on average, 50% of their segregating genes and 100% of their common environment. If similarities within twins (and siblings) arise only from genetic factors, we would expect MZ twin correlations to be (approximately) twice the DZ twin and sibling correlations. If such similarities arise only from shared environmental factors, we would expect MZ twin correlations to be identical to DZ twin and sibling correlations. MZ twin correlations more than twice as large as DZ twin correlations indicate the presence of non-additive genetic factors, which cannot be estimated simultaneously with shared environmental factors in the classical twin design. Therefore, based on a visual inspection of the twin correlations, an A, C, and E or A, D, and E model is chosen. Variation not due to genes or the common environment (i.e., differences between twins) is attributed to the unique environment and measurement error. Similar logic underlies multivariate twin models, in which phenotypic covariances are partitioned into these same components (e.g., A, C, and E). Rather than examining relations between co-twins within a single phenotype (e.g., how well twin 1's honesty-humility relates to twin 2's honesty-humility), this approach examines relations between co-twins across multiple phenotypes (e.g., how well twin 1's honesty-humility relates to twin 2's SDO). Multivariate twin models allow for estimates of the extent to which common sets of genes (genetic correlation, r_g), shared environmental factors (shared environmental correlation, r_c), or unshared environmental factors (unique environmental correlation, r_e) underlie correlations between traits. Importantly, for the variable entered last in the multivariate model, genetic and environmental variance can be partitioned into components shared with all other variables and unique to the variable (see Figure 1(a) and (b)). This approach allows for an estimate of the unique genetic influences on the last variable versus those shared with other variables in the multivariate model.

Before estimating genetic and environmental components of trait variance and covariances, we tested whether twins differed from non-twins and men differed from women in the HEXACO, SDO, and RWA variables. To do so, we first constrained the *means* to be equal across twins and siblings (Model 2) across men and women (Model 3). Second, we constrained *correlations* to be equal for men and women and for sibling and twin members (Model 4 and 5). Within twin members, we tested for quantitative sex differences (i.e., sex differences in heritabilities) by constraining the correlations of the male same-sex MZ twin pairs to be equal to those for female same-sex MZ twin pairs, and the correlations of male DZ twin pairs to be equal to those for female same-sex DZ twin pairs (Model 6). To investigate qualitative sex differences (i.e., sex differences in the genetic and environmental factors underlying variance in men and women), we next constrained the correlation for

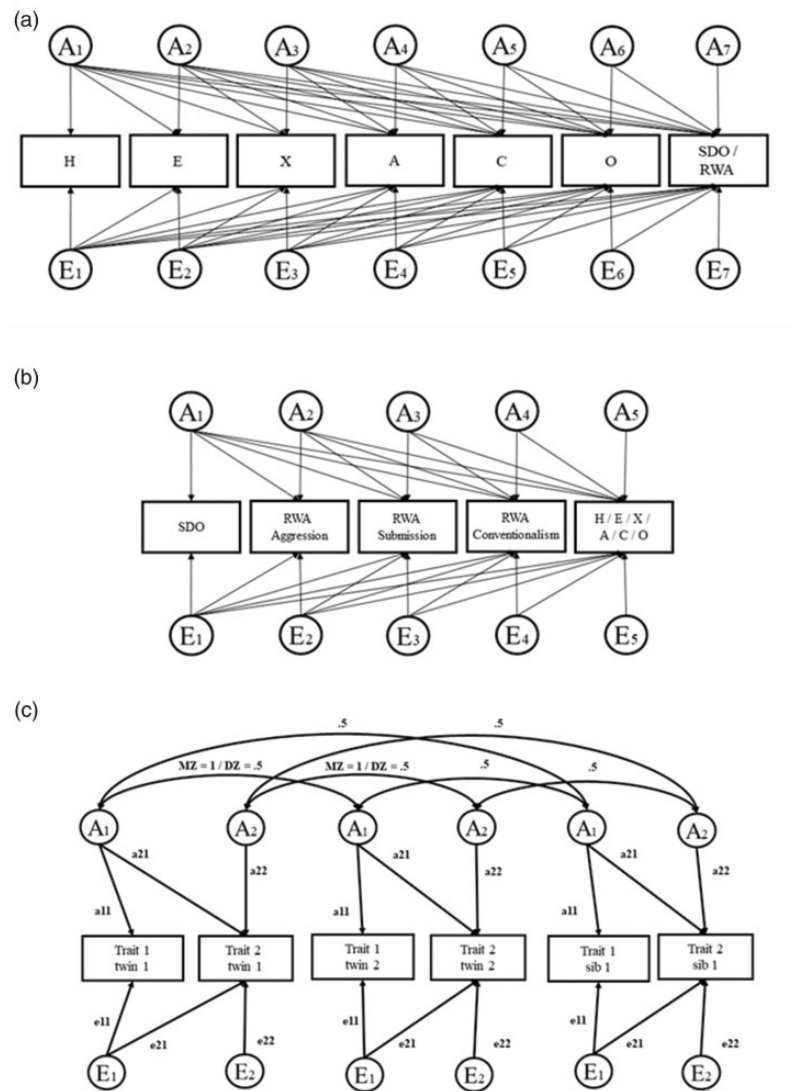


Figure 1. Cholesky models for the multivariate (a and b) and bivariate (c) genetic models. A1 to A7 are additive genetic factors and E to E7 are unshared environmental factors. These models are duplicated for each twin and sibling, one for MZ, one for DZ twins and one for the sibling (see (c) for full model). In (a) and (b), each latent A or E variable can load on the variable beneath (for A) or above (for E) it or to its right, which allowed for a test of the magnitude of unique A and E effects on the last trait of interest entered in the model, or of A and E effects shared with the other variables in the model (for further information, see for instance Neale et al., 1992). MZ: monozygotic; DZ: dizygotic.

DZ same-sex and DOS twins to be equal (Model 7). Lastly, we tested for twin-specific environmental influences by constrained the DZ correlations to be equal to the sibling correlations (Model 8). These analyses (see Supplemental Table S1 for results from all models) revealed sex differences in only the means (with the exception of qualitative sex differences for RWA aggression scale ($p = .004$)) and no evidence for twin-specific environments (meaning that DZ twins were no more similar to each other than non-twin-siblings were to twins). Hence, all analyses described below modelled sex-specific means, but no other sex-specific parameters.

We used likelihood-ratio tests to evaluate constraints on the twin- and sibling correlations and to test whether C or D could be constrained to zero by comparing ACE or ADE models with AE models. In

these tests, the negative log-likelihood ($-2LL$) and the degrees of freedom (df) of the more-constrained model are subtracted from the $-2LL$ and df of the less-constrained model. A chi-square test then shows whether the constraint significantly deteriorates the fit of the model ($p < .01$). We also estimated correlations disattenuated for unreliability by dividing the observed twin correlations by the reliability (i.e., McDonald's omega) separately for each phenotypic correlation. Measurement error-corrected additive genetic (A) and (non-shared) environmental (E) component estimates are provided in Supplemental Table S2.

Table 1. Phenotypic correlations (below the diagonal) with 95% CIs.

	1	2	3	4	5	6	7	8	9	10
1. Honesty-humility	-									
2. Emotionality	.08 (.05, .10)			48% (35–62) 32% (0–64)	39% (6–69)	86% (50–100)	65% (50–78) 76% (36–100)	62% (43–80) 71% (0–100)		
3. eXtraversion	.02 (-.00, .04)	-.11 (-.13, -.09)	41% (10–70)	48% (19–75)	85% (62–100)					
4. Agreeableness	.26 (.24, .28)	-.17 (-.19, -.15)	.13 (.11, .15)	-					93% (65–100)	76% (41–100)
5. Conscientiousness	.14 (.12, .16)	.04 (.02, .06)	.16 (.14, .18)	.07 (.04, .09)	-				65% (16–100)	
6. Openness to experience	.10 (.08, .12)	-.01 (-.03, .02)	.09 (.07, .11)	.08 (.06, .10)	-.01 (-.03, .01)				48% (6–87)	66% (21–100)
7. Social Dominance Orientation	-.30 (-.32, -.28)	-.22 (-.24, -.20)	.01 (-.01, .03)	-.13 (-.15, -.11)	.01 (-.01, .03)	-.27 (-.29, -.25)		84% (73–94)	72% (57–87)	71% (50–91)
8. RWA Aggression	-.22 (-.24, -.20)	-.14 (-.16, -.12)	.01 (-.01, .03)	-.13 (-.15, -.11)	.05 (.03, .07)	-.31 (-.33, -.29)		72% (63–80)	68% (53–83)	73% (58–88)
9. RWA Submission	-.06 (-.08, -.04)	-.03 (-.05, -.01)	.13 (.11, .15)	.10 (.08, .12)	.10 (.08, .12)	-.25 (-.27, -.23)	.51 (.49, .52)	-	66% (50–81)	66% (53–77)
10. RWA Conventionality	-.01 (-.03, .01)	-.01 (-.04, .01)	.12 (.10, .14)	-.00 (-.03, .02)	.10 (.08, .13)	-.18 (-.20, -.16)	.27 (.25, .29)	.27 (.25, .29)	-	60% (45–75)
		2	3	4	5	6	7	8	9	10
McDonalds omega reliability	.82	.84	.90	.82	.79	.83	.84	.70	.70	.76
M	3.63	3.35	3.27	3.06	3.49	3.35	2.62	3.74	2.98	4.25
SD	.56	.58	.67	.54	.50	.62	1.11	1.12	.96	1.01
Additive genetic effects (<i>A</i>)	40% (.34–.46)	34% (.28–.41)	53% (.48–.58)	35% (.29–.59)	38% (.32–.44)	58% (.53–.63)	48% (.42–.53)	46% (.40–.51)	35% (.28–.41)	35% (.29–.41)
Unique A	32%	32%	47%	28%	37%	38%	27%	30%	25%	29%
A shared with SDO, RWA	8 %	2 %	6 %	7 %	1 %	20 %	-	-	-	-
A shared with H E X A C O	-	-	-	-	-	-	21 %	16 %	10 %	6 %
Unique environmental effects (<i>E</i>)	60% (.54–.66)	66% (.59–.71)	47% (.42–.52)	65% (.59–.71)	62% (.56–.68)	42% (.37–.47)	52% (.47–.58)	54% (.49–.60)	65% (.59–.71)	65% (.59–.71)
Unique E	57%	65%	47%	63%	61%	41%	50%	52%	64%	64%
E shared with SDO, RWA	3 %	1 %	0 %	2 %	1 %	1 %	-	-	-	-
E shared with H E X A C O	-	-	-	-	-	-	2 %	2 %	1 %	1 %

Note. The proportion of bolded phenotypic correlations ($r \geq .10$) accounted for by genetic factors is listed above the diagonal. The proportion of the phenotypic correlations explained by unique environmental (E) factors (including the error term) is equal to 100 minus the values above the diagonal. Reliabilities, mean scores (M) and standard deviation (SD) are shown below the correlations. Estimates of A and E effects (plus CI) per phenotype are shown. All best fitting models were AE models. Cross-instrument (personality - ideology) absolute phenotypic correlations $\geq .10$ are bold-faced and only for those the % of genetic effects are shown. HEXACO scales are unit-weighted; Ideology scales are based on factor scores. RWA: Right-Wing Authoritarianism; SDO: Social Dominance Orientation; CI: confidence interval.

Results

Phenotypic correlations between the ideology variables and HEXACO traits

The four ideology variables related more strongly to either Honesty-Humility or Openness to Experience than to the other four HEXACO domain-level variables (Table 1). Relations with honesty-humility were strongest for SDO ($r = -.30$) and RWA aggression ($r = -.22$); they were weak for RWA submission and RWA conventionalism ($r = -.06$ and $-.01$, respectively). Relations with openness to experience were strongest for RWA aggression ($r = -.31$), followed by SDO ($r = -.27$), RWA submission ($r = -.25$), and RWA conventionalism ($r = -.18$). Only one other phenotypic correlation exceeded $|r| = .15$; that between SDO and emotionality ($r = -.22$). Phenotypic correlations between the ideology and personality variables corrected for the remaining personality or ideology variables were generally in the same direction and lower than the zero-order correlations. Results of these corrected correlations are reported in Supplemental Table S3.

Twin correlations

For all variables, MZ correlations were higher than DZ correlations, suggesting that genetic factors underlie phenotypic variance in all traits (see Table 2 for twin and sibling correlations and 95% confidence intervals (CIs); see Supplemental Table S2 for heritability estimates disattenuated for measurement error). DZ correlations were less than half the magnitude of MZ correlations for four of the six HEXACO domain scales and one of the four ideology scales, and they were only slightly larger than half of the magnitude for the remaining variables.

Genetic analyses

In line with a visual inspection of the MZ and DZ correlations, in all twin models, C (the shared environment) or D (non-additive genetic effects) could be constrained to zero (all $p > .05$). That is, we did not detect evidence that common environmental influences made twins and siblings more similar to each other, nor did we detect evidence for non-additive genetic effects. For HEXACO factors, A estimates ranged from .34 for emotionality to .58 for openness to experience. For SDO and the RWA facets, A ranged from .35 for RWA conventionalism and RWA submission to .48 for SDO. At the bottom of Table 1, the unique genetic and environmental effects on the personality and ideology variables are reported together with the genetic and environmental variance in the ideology variables accounted for by genetic and environmental differences in all HEXACO traits on the one hand and the genetic and environmental variance in HEXACO traits accounted for by genetic and

environmental differences in the ideology variables on the other.

The multivariate twin models indicated that the genetic underpinnings of personality varied in their overlap with the genetic underpinnings of SDO and RWA (and vice versa). That is, 8% (H), 2% (E), 6% (X), 7% (A), 1% (C), and 20% (O) of the genetic underpinnings for the HEXACO overlapped with the genetic underpinnings of four ideology variables, and 21% (SDO), 16% (RWA aggression), 10% (RWA submission), and 6% (RWA conventionalism) of the genetic underpinnings of ideology overlapped with those for the six HEXACO variables (Table 1). In Table 3, bivariate genetic and non-shared environmental absolute correlations are reported for phenotypic correlations $\geq .10$ as reported in Table 1. Here, we discuss absolute genetic correlations $\geq .30$. The heritable component of honesty-humility overlapped with the heritable component of SDO ($r_g = -.41$) and RWA aggression ($r_g = -.31$), and the heritable component of openness to experience overlapped with the heritable component of all four ideology variables, i.e., SDO ($r_g = -.49$), RWA aggression ($r_g = -.53$), and RWA submission ($r_g = -.38$). Furthermore, the heritable component of agreeableness overlapped with the heritable component of SDO ($r_g = -.33$) (though we should point out that the phenotypic correlation between these variables was low, $r = -.13$). The highest environmental correlation was between honesty-humility and SDO ($r_e = -.18$).

General discussion and conclusions

This research yields several noteworthy findings. First, it provides an independent test of the factor structure of the recently developed SDO (SDO₇; Ho et al., 2015) and RWA (RWA-ASC; Dunwoody & Funke, 2016) instruments. Findings support the three factors underlying the RWA-ASC, but not for the proposed two-factor structure of the SDO₇. Second, findings provide further evidence for the heritability of both HEXACO traits (average additive genetic variance = .43) and ideology (average additive genetic variance = .41). Third, in line with previous studies, the strongest phenotypic relations are observed between HEXACO honesty-humility and openness to experience and SDO and RWA-ASC. However, compared with previous findings, honesty-humility and openness to experience did not differentially relate to SDO and RWA, respectively. Fourth, findings indicate that more than 60% of the phenotypic relations are explained by genetic factors and that the absolute cross-instrument genetic correlations are generally stronger ($\geq .29$) than the absolute cross-instrument unique environmental correlations ($< .20$). We reflect on each of these findings below.

Table 2. Twin correlations and 95% confidence intervals per zygosity-by-sex group and correlations constrained to be the equal across the sexes (for MZ and DZ twin pairs and siblings) and across DZ twins and sibling.

Correlations												
	Complete pairs	H	E	X	A	C	O	SDO	RWA Aggression	RWA Submission	RWA Conventionalism	
MZm	132	.39 (.24, .52)	.38 (.24, .50)	.58 (.45, .66)	.46 (.32, .57)	.43 (.27, .54)	.59 (.46, .68)	.46 (.31, .57)	.38 (.23, .51)	.36 (.23, .48)	.36 (.19, .49)	
DZm	118	.36 (.21, .49)	.10 (–.08, .29)	.35 (.17, .47)	.24 (.07, .38)	.06 (–.11, .22)	.37 (.20, .50)	.23 (.05, .37)	.28 (.12, .43)	.27 (.11, .40)	.22 (.03, .38)	
MZf	407	.46 (.39, .53)	.41 (.32, .48)	.56 (.50, .63)	.44 (.36, .51)	.45 (.38, .52)	.58 (.52, .64)	.49 (.41, .56)	.49 (.41, .55)	.39 (.30, .47)	.39 (.30, .47)	
DZf	360	.21 (.10, .31)	.19 (.09, .28)	.27 (.19, .37)	.10 (.00, .20)	.22 (.12, .31)	.33 (.25, .41)	.32 (.21, .41)	.34 (.24, .42)	.21 (.10, .31)	.31 (.20, .40)	
DOS	359	.22 (.11, .31)	.07 (–.04, .18)	.19 (.08, .28)	.15 (.05, .25)	.08 (–.02, .19)	.22 (.11, .32)	.19 (.08, .29)	.14 (.03, .24)	.08 (–.02, .19)	.13 (.03, .23)	
MZ	539	.44 (.37, .51)	.40 (.33, .46)	.57 (.51, .62)	.45 (.38, .51)	.45 (.38, .51)	.59 (.52, .64)	.48 (.41, .54)	.46 (.39, .52)	.38 (.31, .45)	.38 (.30, .45)	
DZ	837	.24 (.17, .30)	.13 (.06, .20)	.25 (.18, .31)	.14 (.07, .21)	.15 (.08, .21)	.30 (.23, .36)	.25 (.18, .32)	.25 (.18, .31)	.17 (.10, .24)	.21 (.14, .28)	
Siblings												
Male	272	.09 (–.01, .19)	.12 (.01, .22)	.23 (.12, .33)	.05 (–.05, .16)	.15 (.03, .25)	.36 (.26, .45)	.22 (.12, .32)	.14 (.04, .24)	.07 (–.03, .17)	.05 (–.04, .15)	
Female	606	.22 (.14, .29)	.14 (.07, .21)	.22 (.15, .29)	.09 (.01, .17)	.12 (.05, .19)	.25 (.18, .32)	.22 (.14, .28)	.24 (.17, .31)	.16 (.08, .24)	.15 (.08, .22)	
N siblings	878	.18 (.12, .25)	.13 (.06, .19)	.23 (.17, .29)	.08 (.01, .15)	.13 (.05, .18)	.27 (.21, .32)	.22 (.15, .28)	.21 (.15, .27)	.13 (.06, .19)	.12 (.05, .18)	
N sibs + DZ	1715	.20 (.16, .26)	.13 (.08, .19)	.24 (.19, .28)	.11 (.06, .16)	.14 (.08, .18)	.29 (.23, .32)	.23 (.18, .28)	.22 (.18, .28)	.15 (.10, .19)	.15 (.10, .20)	
M												
Male	350	3.50	3.58	3.27	3.19	3.47	3.37	3.00	4.00	3.08	4.32	
Female	368	3.68	2.93	3.27	3.01	3.51	3.30	2.41	3.58	2.97	4.19	
SD		0.55	0.50	0.67	0.54	0.50	0.62	1.06	1.09	0.95	1.01	

Note. Mean scores and standard deviations are depicted at the bottom of the table. HEXACO scales are unit-weighted; Ideology scales are based on factor scores. RWA: Right-Wing Authoritarianism; SDO: Social Dominance Orientation; MZm: monozygotic male; DZm: dizygotic male; MZf: monozygotic female; DZf: dizygotic female; DOS: dizygotic opposite-sex; MZ: monozygotic; DZ: dizygotic.

Table 3. Genetic correlations (below the diagonal) and unique environmental correlations (above the diagonal) for the most parsimonious AE models.

	H	E	X	A	C	O	SDO	RWA Aggression	RWA Submission	RWA Conventionalism
Honesty-Humility	—									
Emotionality		—								
eXtraversion			—							
Agreeableness	.42 (.31, .52)	-.12 (-.22, -.03)	-.14 (-.20, -.08)	.27 (.20, .33)	.13 (.06, .20)	.03 (.00, .10)	-.18 (-.24, -.12)	-.14 (-.21, -.08)	.02 (-.04, .08)	.05 (-.01, .12)
Conscientiousness	.13 (.02, .24)	-.12 (-.24, .01)	.16 (.06, .25)	.13 (.06, .20)	.05 (-.02, .11)		-.04 (-.11, .02)	-.02 (-.08, .04)	.05 (-.02, .11)	
Openness to exper:	.18 (.09, .27)		.31 (.22, .40)	—	—		-.06 (-.13, .01)	-.08 (-.14, .01)	.08 (.02, .15)	.05 (-.01, .12)
SDO	-.43 (-.51, -.34)	-.19 (-.29, .08)		-.33 (-.43, -.23)		-.49 (-.56, -.42)	-.10 (-.16, .03)	-.11 (-.17, .04)	-.13 (-.19, -.06)	-.10 (-.17, -.03)
RWA aggression	-.31 (-.40, -.22)	-.08 (-.18, .03)		-.26 (-.37, -.16)		-.53 (-.60, -.47)	—	.25 (.19, .32)	.15 (.08, .21)	.12 (.05, .18)
RWA submission			.29 (.19, .39)	.16 (.04, .29)	.13 (.02, .25)	-.38 (-.46, -.30)	.74 (.67, .81)	—	.14 (.07, .21)	.19 (.12, .25)
RWA convention.			.20 (.10, .29)		.17 (.05, .29)	-.29 (-.38, -.20)	.44 (.34, .53)	.41 (.31, .51)	—	.17 (.11, .23)
							.47 (.37, .57)	.54 (.44, .64)	.48 (.36, .59)	—

Note. HEXACO scales are unit-weighted; Ideology scales are based on factor scores. Only genetic and unique environmental correlations that are based on at least a phenotypic $|r| \geq .10$ (see Table 1) are reported. Cross-instrument genetic and environmental $|r| \geq .30$ are bold-faced. RWA: Right-Wing Authoritarianism; SDO: Social Dominance Orientation.

Factor structure of SDO and RWA

Although both SDO and RWA have been argued to be multi-faceted (Dunwoody & Funke, 2016; Ho et al., 2015), findings here support the multi-faceted nature of RWA, but not of SDO. With respect to SDO, the SDO₇ is supposed to contain two facets that respectively represent *blatant* (SDO-D) and more *subtle* (SDO-E) forms of group inequality support. One of the possible reasons for the strong relation between SDO-D and SDO-E observed in this study may be that blatant SDO-D fully subsumes subtle SDO-E. That is, if respondents indicate that one group should dominate other groups, they are in fact implying that groups should not be equal and, thus, scoring high on SDO-E is a necessary—albeit maybe not sufficient—condition for scoring high on SDO-D. Consequently, the distinction between the endorsement of the ideological content in the SDO-D and SDO-E scales may be rather a matter of degree than of content, which in turn may imply that item-response theory (IRT) models are more suitable to examine distinctions between SDO-D and SDO-E.

In contrast, we found clear differences between three facets of RWA, (authoritarian) aggression, (authoritarian) submission, and conventionalism, which correlate (phenotypically) on average .30 with each other. In terms of item content, the three facets are clearly distinct, referring respectively to the endorsement of violence by authorities (aggression), the uncritical believe in authorities (submission), and the importance of norms and traditions (conventionalism). That is, none of these three facets can be seen as a *sine qua non* for the other facets. Notably, the (authoritarian) aggression facet is more closely related to SDO than it is to the two other RWA facets. This close association may reflect RWA aggression's reference to the use of force against "threatening groups." High SDO individuals prefer their group(s) to enforce inequality/dominance over other groups. Thus, both RWA aggression and SDO share a positive attitude towards using force to maintain group-based power differences. It is also of note that HEXACO honesty-humility more strongly related to SDO and RWA aggression than to RWA submission and RWA conventionalism, implying that the former two share content that the latter two do not share to the same degree. In total, these findings suggest that two-dimensional models of ideology may be better captured by a combination of SDO dominance/inequality and RWA aggression and a combination of RWA submission and RWA conventionalism than by SDO and an aggregate of RWA-ASC.

Heritability of HEXACO personality and ideology variables

In line with previous studies, we found no evidence for shared environmental effects on HEXACO personality dimensions. In contrast to other findings (Funk et al., 2013; Kandler et al., 2016; McCourt et al., 1999), we also found no shared environmental effects for the SDO and RWA variables (estimates of A and E components are listed in Table 1 and Supplemental Table S2). Our findings are, however, consistent with those reported in another recent study (Kleppestø et al., 2019), which reported SDO heritability estimates similar to that reported here (.37 for SDO-D and .24 for SDO-E, respectively). The contrasts in findings are hard to explain by looking at sample-specific differences. That is, all samples are drawn from Western countries (e.g., the US in Funk et al. (2013); mainly US- and UK-born in McCourt et al. (1999); Germany in Kandler et al. (2016); Norway in Kleppestø et al. (2019), and Finland in this study), with reasonably similar sample sizes. For now, we can only speculate about the reasons for the difference in heritability and shared environmental estimates in the ideology variables; more research is needed to resolve this issue.⁷

As is the case in most twin studies of personality and/or ideology, heritability estimates reported here are attenuated by measurement error. As a consequence, uncorrected heritability estimates should be interpreted as a lower bound of the “true” estimate. In Supplemental Table S2, we provide not only the AE estimates, but also estimates corrected for attenuation. The results show that the disattenuated heritability estimates are higher than their attenuated counterparts. The average disattenuated additive genetic estimates are .52 for the HEXACO traits and .55 for the ideology variables. However, note that correcting for unreliability can lead to overcorrection (but see De Vries (2013), for an empirical example using the HEXACO model in which overcorrection is minimal), and thus corrected heritability estimates may need to be interpreted as providing an upper bound of heritability estimates (Briley & Tucker-Drob, 2014).

Phenotypic relations HEXACO personality and ideology variables

Although honesty-humility and openness to experience related most strongly to SDO and RWA dimensions, relations were not as clear-cut as in some previous studies, which suggested that SDO relates mainly to honesty-humility and RWA mainly to openness to experience (Lee et al., 2010). Honesty-humility did more clearly distinguish between the two by relating more strongly to SDO and RWA aggression than to RWA submission and RWA conventionalism. Although the difference in relations

between honesty-humility and SDO on the one hand and between honesty-humility and these latter two RWA-ASC variables on the other hand are in line with earlier studies, the much more equal relations between openness to experience and SDO and RWA are not. As noted in the introduction, people high in openness to experience are more likely to seek out situations that allow for exploration. In situations in which groups are more unequal and separated (i.e., in a high social dominance society), it would be more difficult for those high on openness to experience to get in contact with other groups (situation activation) and to display and satisfy their curiosity (trait and outcome activation; De Vries et al., 2016). That is, inequality of groups may hamper the activation and expression of openness to experience.

However, it should be noted that in seven out of eight studies reported across four articles (Desimoni & Leone, 2014; Lee et al., 2010; Leone et al., 2012a; Leone et al., 2012b), relations between openness to experience and RWA were stronger than relations between openness to experience and SDO (the exception being Study 2 in Leone et al., 2012b). All of these studies used older instruments to measure SDO and RWA, which may explain the differences in results. Thus, generally, care should be taken by interpreting the findings of our study that are in contrast with earlier studies. Clearly, more research is needed with the SDO₇ to establish whether these divergent findings (with respect to the factor structure and relations with HEXACO traits) are substantive, due to sample-specific fluctuations, or due to differential instrument-specific content.

Perhaps most notably, the current study provides estimates of the degree to which phenotypic correlations between HEXACO personality and ideology variables are explained by genetic versus environmental factors. Genetic factors accounted for more than 60% of every phenotypic correlation greater than $|r| = .10$. Genetic factors also accounted for more than 60% of correlations between the four ideology variables. Consequently, environmental factors (all of which were not shared by twins) accounted for no more than 40% of these phenotypic correlations.

Genetic and environmental correlations between HEXACO personality and ideology variables

We detected sizable overlap in the genetic factors (with $|r_g|$'s $> .30$) underlying (a) honesty-humility, SDO, and RWA aggression and (b) openness to experience and all ideology scales except RWA conventionalism. The only other genetic correlation greater than $|r_g| = .30$ was that between agreeableness and SDO (though we highlight that, in contrast with the relation between SDO and Big Five agreeableness, the phenotypic correlation between the variables was weak). Only a few environmental correlations had 95% CIs that did not overlap with zero, and those

CIs did not overlap with the point estimates for the genetic correlations. Ultimately, then, results suggest that there is greater overlap in the genetic underpinnings of personality and ideology than there is in the environmental underpinnings of personality and ideology.

Findings here are especially notable in light of recent work reporting that genetic variance tends to decline for most personality variables, especially in the latter part of life (Kandler et al., 2021), with idiosyncratic environmental factors playing an increasingly important role. The decline in heritability should—one would think—provide more room for life experiences to affect both personality and ideology, which would show itself through higher environmental variance and higher environmental correlations between personality and ideology. Recent research has looked at such personality changes, especially with respect to honesty-humility and openness to experience (Thielmann and De Vries, 2021), the two key variables in our study. Relatively large changes were observed in honesty-humility and openness to experience, but these changes could not be explained through environmental factors, nor through volitional change factors.

Can we conclude, based on the findings, that the ideology variables should be considered personality variables? The question is partly a semantic one and may depend on the definitions scholars have of personality and ideology or values. Based on previous research, RWA has been found to have substantial heritability and (heritable) stability (Ludeke & Krueger, 2013), which is in line with common definitions and conceptualizations of personality. This research not only shows that RWA and SDO have substantial heritability, but also that the relations between personality and these two ideology variables are mostly genetic in origin. That is, there is sufficient (genetic) overlap between personality and ideology to consider the latter at least partly as personality traits. More research needs to be done to show whether the ideology variables also share variance with variables that lie outside the personality domain and/or whether the genetic or environmental variance in (HEXACO) personality and ideology (or values) variables explain the same or different outcomes.

Limitations and future research

Standard limitations of classical twin modeling apply here (Verweij et al., 2012). Most importantly, valid interpretation of the results rests upon the equal environment assumption, which states that MZ twins and DZ twins experience equally correlated environments. Additionally, DZ twins are assumed to share on average 50% of their segregating genes. This assumption is only valid if mating occurs at random with respect to the phenotype(s) being investigated, and it is hence violated in the presence of assortative mating or

inbreeding. Because C and D cannot be estimated simultaneously, only A and C or A and D can be modeled, which can lead to an underestimation of C and/or D influences and an overestimation of A influences. Lastly, interactions or correlations between trait-relevant genes and environmental influences can inflate A, C, or E estimates. Some of these limitations are addressed by more sophisticated models that allow for the estimation of sources of variance that are impossible to disentangle using classical twin designs. For instance, extended twin family designs, such as the Nuclear Twin Family Model (NTFM; Keller et al., 2010; for examples, see Kandler et al., 2016; Kandler et al., 2019), which adds parent self-ratings to the design, allows for the estimation of passive nature-nurture effects and separate parental effects. Adding spouses to the design in a Spouses-of-Twins Model (SoTM; Kandler et al., 2015; Kandler et al., 2019) allows for the estimation of the effects of shared social background (social homogamy) and spouse-specific interactions. Peer ratings might also be a useful addition, given that they can be combined to yield more accurate estimates of the systematic variance in each of the variables, which may help to obtain more accurate heritability and environmental estimates (Kandler et al., 2016). In turn, employing such designs may also further our understanding on more specific sources of genetic, shared, and unshared environmental variance that are responsible for bivariate relations. For instance, social homogamy effects observed in honesty-humility (Kandler et al. 2019) and SDO (Kandler et al., 2015) may (partly) drive the relation between the two variables, something that future research may like to investigate.

The current study is also limited by its cross-sectional design. Although causality is difficult (and, perhaps, impossible) to assess using nonexperimental designs (i.e., due to the possible existence of extraneous variables), longitudinal data may go some way in answering the question whether changes in personality drive changes in ideology or vice versa. Although non-genetic longitudinal studies have provided some evidence for the existence of causal relations from personality to ideology (Perry & Sibley, 2012; Sibley & Duckitt, 2013), direction-of-causality models (e.g., Duffy & Martin, 1994) based on twin studies have not been able to confirm such causal claims, showing that the relations between personality and ideology are more likely to have a common genetic cause (e.g., pleiotropy; Kandler et al., 2014; Verhulst et al., 2012). Future studies might like to further investigate this question by using a longitudinal design to test whether genetic or environmental effects drive longitudinal changes in personality and ideology variables.

An asset of this study is that it used the latest—and full—versions of the SDO and RWA-ASC scales. Both of them have been constructed using pro- and con-trait items and both of them have been

thoroughly tested (Dunwoody & Funke, 2016; Ho et al., 2015). Interestingly, these new instruments may also be the cause of the strong SDO–RWA aggression relation and the—when compared to other studies—different relations between HEXACO honesty-humility and openness to experience with SDO and RWA. Clearly, more research needs to be done using these new instruments to see if the current findings replicate.

Future research might also focus on the genetic relations between HEXACO traits, SDO, and RWA and generic—circumplex—values (Schwartz, 1992) and social attitudes (Saucier, 2000). It would be especially interesting to find out whether RWA aggression more strongly relates (both phenotypically and genetically) to values associated with self-enhancement versus self-transcendence than with those related to openness to change versus conservatism. Integrating personality, the value circumplex, and socio-political attitudes and ideology in one overarching model will be an important step towards the unification of psychological, sociological, political, and environmental research efforts in this domain.

To conclude, this study shows support for a shared genetic background of personality—mainly honesty-humility and openness to experience—and socio-political ideology. The findings of our study align with previous heritability estimates of .40 to .50 for personality and add to previous research by showing a genetic overlap—and only weak environmental overlap—of personality and ideology. Our study showed that SDO, like RWA, has a strong heritable component and almost no shared environmental component. It also deviates from other studies (e.g., Ho et al., 2015) in showing that the two SDO facets are virtually indistinguishable. Furthermore, our research showed a strong (phenotypic and genetic) overlap between SDO and RWA aggression, and also expressed through their convergent associations with honesty-humility. All in all, the findings show that ideology is partially grounded in the heritable part of personality, and that honesty-humility and openness to experience are the two most important personality dimensions that underlie individual differences in ideology that have been shown—in previous studies—to be related to how we vote, how we relate to our environment, and how we treat other ethnic groups.

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Data accessibility statement

All data and analysis scripts that underlie the findings of this study are openly available at <https://osf.io/yhxwt/>.

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ORCID iD

Reinout E. de Vries  <https://orcid.org/0000-0002-4252-5839>

Notes

1. Honesty-humility shares some variance with Big Five agreeableness, but Big Five agreeableness variance is also redistributed among HEXACO agreeableness and HEXACO emotionality, such that anger, which is part of Big Five (reversed) emotional stability, is part of HEXACO agreeableness, and sentimentality, which is part of Big Five agreeableness, is part of HEXACO emotionality.
2. Note that Kam et al. (2013), Lewis and Bates (2014b), Veselka et al. (2009), and Veselka, Petrides, et al. (2010) used a (subset of) the same UK twin sample used by Veselka, Schermer, et al. (2010).
3. The composite heritability estimates of self- and peer reports in Kandler et al. (2016)—and not the true score heritability estimates—are reported here. Furthermore, these estimates were based on scores that were corrected for the common variance shared with the other ideology variable. The uncorrected heritability estimates in the same data were higher (e.g., .57 for RWA and .20 for SDO; see Kandler, 2015). In what follows, the composite shared environmental estimates of Kandler et al. (2016) are reported.
4. Readers interested in the phenotypic and genetic relations between other individual difference (e.g., cognitive ability) and personality (e.g., psychoticism) variables and other conceptualizations of ideology (e.g., liberal political attitudes) are referred to Hatemi and Verhulst (2015), Oskarsson et al. (2015), Verhulst et al. (2010), and Verhulst et al. (2012). For instance, Verhulst et al. (2010, 2012) showed that psychoticism was negatively related to—and shared substantial genetic variance with—liberal political attitudes. In order to translate the findings of psychoticism to the HEXACO model, Dunlop et al. (2012) showed that the most important (negative) correlates of psychoticism are honesty-humility, emotionality, and conscientiousness. These findings, although not one-on-one translatable to the HEXACO model, may provide some further indication that openness to experience (being positively related to cognitive ability; DeYoung et al., 2014) and honesty-humility (being negatively related to psychoticism) are among the main drivers of the personality–ideology relations.

5. The current sample has also been used in Kandler et al. (2021); however, this particular study used only a subset of HEXACO items (i.e., the HEXACO-60 instead of the HEXACO-100) and does not report on phenotypic, genetic, and environmental relations with SDO and RWA.
6. We also performed a CFA on the HEXACO items, with items loading on their designated facets, and facets loading on their designated factors. The results are reported on <https://osf.io/yhxwt/>. Although traditional fit estimates were less than adequate ($\chi^2(4560) = 244,123.73$, $p < .001$, CFI = .74, TLI = .73), which is understandable for such a complex broad bandwidth personality models, the RMSEA was adequate at .04 [90% CI: .04, .04], all items loaded significantly on their designated facets, and all facets loaded significantly (with all standardized loadings $> .50$) on their designated factors. In line with common practice, we used the unit-weighted factor scores to measure the six HEXACO domain scales.
7. Assortative mating may have played a role in the significant shared environmental effect, observed in some of the studies (Funk et al., 2013; Kandler et al., 2016; McCourt et al., 1999). Note that Kandler et al. (2016) obtained heritability estimates for both self- and peer reports. For SDO, these estimates were almost identical (.05 for self-reports and .01 for peer reports); however, for RWA, the heritability estimates were considerably lower for peer reports (.20) than for self-reports (.54). Response styles that have been found to be partially heritable as well (Melcher et al., 2018), may account for some of the differences between the findings in self- and peer reports.

Supplemental material

Supplemental material for this article is available online.

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